Detection and Risk Assessment of Unsafe Hanging Travel in Public Transport Using Machine Learning

Tazkia Malik, Sadek Hossain Asif, Tanzila Khan, Taif Ahmed Turjo, Amio Rashid, Mahmud Hasan Walid
Department of Computer Science and Engineering
University of Dhaka
Dhaka, Bangladesh

***Abstract*—Unsafe door-hanging and overcrowded travel are persistent safety risks in Dhaka's public transport system, particularly in buses and legunas operating under high passenger demand. Existing intelligent transportation research has addressed vehicle detection, passenger counting, and general crowd analysis, but localized automated detection of unsafe hanging travel remains limited. This paper presents an image-based machine-learning framework for classifying public transport scenes into safe/normal and unsafe/risky conditions. A custom bus and leguna image dataset is prepared through cleaning, annotation, preprocessing, and augmentation. For classical machine-learning baselines, Histogram of Oriented Gradients (HOG) descriptors combined with Logistic Regression, Gaussian Naive Bayes, and a Support Vector Machine (SVM) using a radial basis function kernel. Deep models including a CNN trained from scratch, ResNet18, EfficientNet-B0, ResNet50, ConvNeXt-Tiny, and an ensemble are also evaluated to compare learned visual features with handcrafted HOG representations. The retrained dataset contains 523 annotation-labeled images, and augmentation is restricted to the training partition to avoid leakage. On the untouched 79-image test set, the ResNet50–ConvNeXt ensemble reaches 94.7% unsafe recall, 86.1% accuracy, and 0.970 ROC-AUC at the balanced operating point, while ResNet50 attains the highest single-model accuracy of 92.4%. Because a missed hanging passenger is more costly than a false alarm, the system is tuned to favor unsafe-class recall. The results indicate that a localized, annotation-driven dataset with leakage-free augmentation allows standard transfer-learning models to detect unsafe hanging travel with high reliability.**

***Index Terms*—Unsafe public transport, computer vision, image classification, Histogram of Oriented Gradients, support vector machine, convolutional neural network, intelligent transportation system.**

I. Introduction

Public transport in Dhaka carries a large share of daily urban mobility, but high passenger demand, limited vehicle capacity, and inconsistent enforcement often create unsafe conditions. A common visible hazard is door-hanging travel, where passengers ride from the doorway or exterior boundary of buses and legunas, increasing exposure to falls, collisions, and roadside obstacles.

Safety monitoring still depends heavily on traffic personnel, transport operators, or surveillance staff. Manual observation is difficult to scale across dense road networks, heterogeneous vehicle types, and continuously changing traffic scenes.

Computer vision is useful for this task because images can capture vehicle structure, passenger position, crowding, and doorway conditions. This paper proposes an image-based machine-learning system that classifies Dhaka public transport scenes into safe/normal negative samples and unsafe/risky positive samples.

The main contributions are as follows:

- A localized safe/unsafe public transport image dataset containing Dhaka bus and leguna scenes.
- A complete machine-learning pipeline for unsafe hanging-travel detection.
- A comparison of HOG-based classical machine-learning models and deep convolutional models.
- An evaluation using standard classification metrics, including accuracy, unsafe-class recall, ROC and precision–recall curves, and AUC, together with a train/validation/test generalization analysis.

II. Related Work

Computer vision has become an important component of intelligent transportation systems. Surveyed applications include vehicle detection, pedestrian detection, anomaly detection, traffic monitoring, and automatic number-plate recognition [1]. These methods show that visual data can support automated interpretation of transportation scenes, although most work targets general traffic understanding rather than specific passenger-safety risks.

Passenger monitoring research commonly focuses on counting, occupancy estimation, boarding analysis, and crowd monitoring in buses or terminals [2]. Public transport safety studies identify overcrowding, weak enforcement, and unsafe passenger behavior as important risk factors in dense urban environments [3], while broader reviews show that crowd pressure and non-collision incidents remain persistent concerns when demand exceeds service capacity [4].

Unsafe behavior and abnormal-activity detection are related to the present task because they aim to identify visual patterns that deviate from normal scenes. Recent surveys report that deep learning can be effective for abnormal behavior detection, but they also emphasize difficulties caused by limited labeled data, occlusion, scene variation, and domain transfer [5]. Object-detection methods such as YOLO provide another foundation for transportation surveillance and boarding or alighting recognition [6], [7]. Closer to the present task, vision-based systems have been developed to recognize abnormal or unsafe passenger behavior inside buses from onboard cameras [8], [9], [10], and to analyze passenger posture during the

TABLE I
DATASET COMPOSITION AFTER SPLITTING AND TRAIN-ONLY AUGMENTATION

Partition	Total	Safe	Unsafe
Training (augmented)	2,730	1,330	1,400
Validation	79	41	38
Test	79	41	38
Labeled originals	523	272	251

safety-critical boarding and alighting phase at bus doors [11]. These efforts confirm that onboard and door-region imagery carries strong cues for passenger-safety assessment, but they target in-vehicle activity or door movement rather than passengers hanging from the exterior of a moving vehicle.

Automatic License Plate Recognition (ALPR) is relevant only as future context for identifying vehicles after unsafe behavior is detected. Existing ALPR studies address plate localization and recognition under real-world imaging constraints [12], [13], [14], including Bengali license-plate recognition under low-resolution and blurred conditions [15].

The key gap is that existing work rarely addresses unsafe door-hanging detection in Dhaka-like crowded public transport environments using localized datasets under real-world challenges such as occlusion, motion blur, lighting variation, crowding, and non-standard vehicle structures. Prior systems often estimate occupancy, count passengers, or recognize boarding events, but they do not directly model hanging passengers as a safety-critical positive class under local bus and leguna conditions. Safety interventions for footboard travel in South Asian buses have relied mainly on onboard IoT sensors rather than vision [17], and although localized Bangladeshi image datasets exist for vehicle-type classification [16], none label unsafe door-hanging behavior.

III. DATASET

A. Data Collection

The dataset contains images of buses and legunas captured under real-world public transport conditions in Dhaka. The samples cover safe/normal and unsafe/risky cases with variation in vehicle type, passenger density, camera angle, lighting, and scene clutter. Labels are derived from image annotations: an image is unsafe if it contains at least one box labeled `unsafe`; otherwise it is safe, and license-plate boxes are ignored. The retrained dataset contains 523 labeled originals, with 272 safe samples and 251 unsafe samples. Figure 1 summarizes the updated class distribution and split composition, and Table I lists the per-partition counts.

Representative examples of the vehicle categories are shown in Fig. 2 and Fig. 3.

B. Data Cleaning

Cleaning removed images whose content was blurred, ambiguous, low quality, heavily obstructed, poorly framed, or unreliable for class assignment. This step reduced label noise while preserving realistic variation in traffic, vehicle structure, crowding, and illumination.

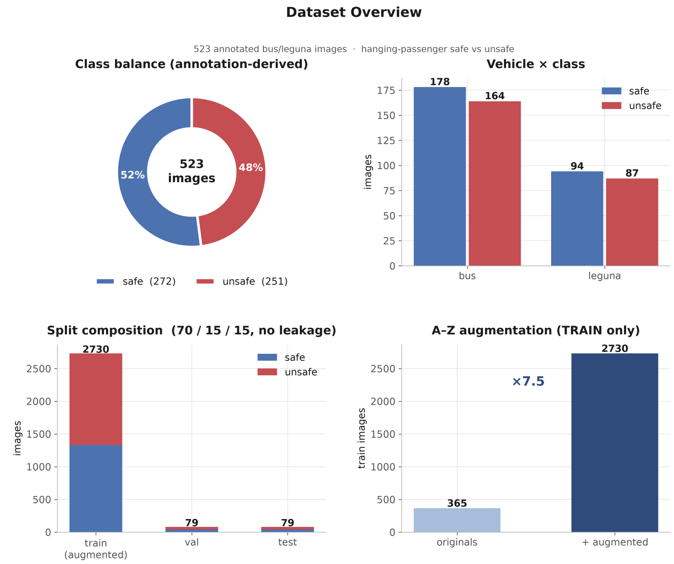


Fig. 1. Dataset overview: annotation-derived class balance, vehicle-wise class distribution across buses and legunas, stratified 70/15/15 split composition, and train-only augmentation growth from 365 originals to 2,730 training images.

(a) Negative class.

(b) Positive class.

Fig. 2. Representative leguna samples.

C. Data Annotation

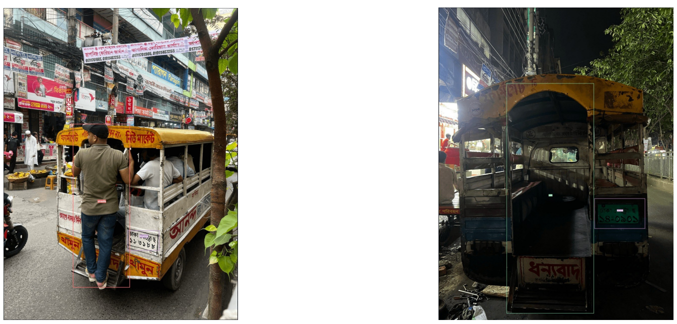
Manual annotation assigned each retained image to one of two classes: negative for safe/normal passenger conditions and positive for unsafe/risky hanging travel. Figure 4 shows representative annotation examples.

D. Data Augmentation

Augmentation was restricted to the training partition to prevent leakage into validation or test data. The 365 training originals produced 2,365 offline augmented samples, yielding 2,730 training images with 1,330 safe and 1,400 unsafe samples. Validation and test each contain 79 unaugmented originals, divided into 41 safe and 38 unsafe images. Deep models additionally used online augmentation, weighted sampling, and horizontal-flip test-time augmentation. These transformations simulate common roadside imaging variation while preserving the class label [18].

(a) Negative class. (b) Positive class.

Fig. 3. Representative bus samples.



(a) Unsafe/risky annotation. (b) Safe/normal annotation.

Fig. 4. Representative annotation examples.

All images were standardized before model-specific processing. The standardized preprocessing and model input settings are summarized in Table II. The dataset is available online at https://drive.google.com/drive/folders/1D3jDBvA-cAhwRPA4wUBUss5zlwdXdLJY.

E. Ethical Considerations

The dataset is used only for academic safety-monitoring research. Images may contain privacy-sensitive visual information such as human faces, vehicle license plates, and location-related context. The classification task focuses on transport safety conditions rather than personal identity recognition; no face recognition, identity matching, or individual profiling is performed. Before any public release or operational deployment, privacy-preserving steps such as face and license-plate anonymization, removal of GPS or location metadata, consent and data-governance procedures, and compliance with relevant regulations would be required.

IV. Methodology

The proposed method is a binary image-classification pipeline for public transport safety assessment. The workflow consists of dataset preparation, preprocessing, feature extraction or model-specific input preparation, model training, and evaluation. HOG-based classical models receive handcrafted gradient features, while deep neural models learn directly from image tensors. Implementation settings are summarized in Table II.

A. Pipeline Overview

The pipeline begins with Dhaka bus and leguna image collection, followed by cleaning, annotation, augmentation, and standardization. Classical machine-learning models use HOG feature vectors extracted from grayscale inputs, whereas CNN and transfer-learning models receive RGB tensors. All models are evaluated on the same held-out split using standard classification metrics.

B. Training and Inference Procedure

The complete retraining procedure is summarized below so that data preparation, model fitting, and deployment decisions are defined within the methodology.

1) **Annotation-based labeling:** Begin with 523 labeled originals containing 272 safe and 251 unsafe images. Assign an image to the unsafe class when at least one annotation box is labeled `unsafe`; ignore license-plate boxes.
2) **Four-way stratification:** Split the originals 70/15/15 over vehicle type and class, preserving bus-safe, bus-unsafe, leguna-safe, and leguna-unsafe samples in validation and test partitions.
3) **Leakage-free augmentation:** Apply offline augmentation only to the 365 training originals, producing 2,730 training images. Keep the 79-image validation and 79-image test partitions unaugmented.
4) **Model-specific preparation:** Extract standardized HOG features followed by PCA for classical models. Supply normalized RGB tensors to the scratch CNN and ImageNet-pretrained transfer-learning models.
5) **Model training:** Train Logistic Regression, Gaussian Naive Bayes, SVM, and the scratch CNN as baselines. Fine-tune ResNet18, EfficientNet-B0, ResNet50, and ConvNeXt-Tiny in two phases. Use weighted sampling and class-weighted loss for deep models.
6) **Validation and ensembling:** Tune balanced, high-recall, and max-recall decision thresholds on validation predictions. Form the final ensemble by averaging ResNet50 and ConvNeXt-Tiny unsafe-class probabilities.
7) **Test-time inference:** Apply horizontal-flip test-time augmentation to the deep models, average the resulting probabilities, select the configured operating threshold, and evaluate once on the untouched test set.

C. HOG-Based Feature Representation

For the classical models, Histogram of Oriented Gradients (HOG) descriptors were extracted from grayscale images [19]. HOG captures local gradient orientation and edge-structure information, which is useful for posture-, shape-,

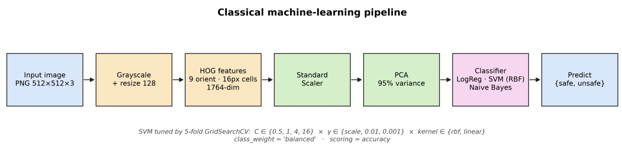


Fig. 5. Overview of the HOG-based classical machine-learning pipeline.

and boundary-related visual classification. Feature extraction was implemented with scikit-image [20]; Table II lists the extraction settings, and Fig. 5 shows the full classical pipeline.

D. Classical Machine Learning Models

The classical baselines use HOG features with three complementary classifiers. Logistic Regression provides a linear probabilistic baseline and estimates the posterior probability of the positive (unsafe) class as

$$P(y = 1 \mid \mathbf{x}) = \sigma\big(\mathbf{w}^\top \mathbf{x} + b\big) = \frac{1}{1 + \exp(-(\mathbf{w}^\top \mathbf{x} + b))}, \tag{1}$$

where $\mathbf{x} \in \mathbb{R}^d$ is the PCA-reduced HOG feature vector, $\mathbf{w} \in \mathbb{R}^d$ and $b \in \mathbb{R}$ are the learned weight vector and bias, and $\sigma(\cdot)$ is the logistic sigmoid. The model is computationally efficient and interpretable, but its linear decision boundary can be limited when safe and unsafe samples are not linearly separable.

Gaussian Naive Bayes is a probabilistic baseline that assumes conditional independence among features given the class label. Each feature x_i is modeled with a class-conditional Gaussian likelihood,

$$\begin{aligned} p(x_i \mid y) &= \mathcal{N}\big(x_i;\, \mu_{y,i},\, \sigma_{y,i}^2\big) \\ &= \frac{1}{\sqrt{2\pi\, \sigma_{y,i}^2}} \exp\left(-\frac{(x_i - \mu_{y,i})^2}{2\,\sigma_{y,i}^2}\right), \end{aligned} \tag{2}$$

where $\mu_{y,i}$ and $\sigma_{y,i}^2$ are the mean and variance of feature i under class y. Assuming conditional independence, the predicted label maximizes the class posterior,

$$\hat{y} = \underset{y \in \{0,1\}}{\arg\max}\; P(y) \prod_{i=1}^{d} p(x_i \mid y). \tag{3}$$

Although the independence assumption is restrictive for correlated HOG features, the model is useful as a low-complexity comparator. The Support Vector Machine (SVM) identifies a decision boundary that maximizes the margin between the negative and positive classes [21]. To capture non-linear patterns in the HOG feature space, the radial basis function (RBF) kernel is used to quantify the similarity between pairs of feature vectors,

$$K(\mathbf{x}_i, \mathbf{x}_j) = \exp\big(-\gamma\, \|\mathbf{x}_i - \mathbf{x}_j\|_2^2\big)\,, \qquad \gamma > 0, \tag{4}$$

and the decision function for a test vector $\mathbf{x}$ is

$$f(\mathbf{x}) = \operatorname{sgn}\left(\sum_{i \in \mathcal{S}} \alpha_i\, y_i\, K(\mathbf{x}_i, \mathbf{x}) + b\right), \tag{5}$$

where $\mathcal{S}$ indexes the support vectors, $\alpha_i \geq 0$ are the learned dual coefficients, $y_i \in \{-1, +1\}$ are the training labels, and b is the bias. The penalty C, kernel width γ, and kernel type were selected by five-fold grid search on the training data. Table II summarizes the implementation settings for all three classical models.

E. Deep Neural Models

The deep models process RGB images directly and learn spatial features from the training data [22]. The experiments evaluate five neural architectures: a CNN trained from scratch, ResNet18, EfficientNet-B0, ResNet50, and ConvNeXt-Tiny. The scratch CNN provides a lightweight learned-feature baseline, while the other architectures use ImageNet-pretrained weights and two-phase fine-tuning [23], [24], [25]. A probability-level ensemble combines ResNet50 and ConvNeXt-Tiny. All neural models use batch normalization [26] for stable optimization and dropout [27] for regularization. Weighted sampling, class-weighted loss, validation-tuned thresholds, and horizontal-flip test-time augmentation address imbalance and improve unsafe-class recall. Figure 6 shows the scratch CNN architecture and Fig. 7 illustrates the two-phase transfer-learning scheme shared by the pretrained models; Table II reports the combined configuration.

V. Experimental Setup

A. Split Protocol

The data split followed a four-way stratified protocol over vehicle type and class, producing 70/15/15 training, validation, and test partitions. This ensures that bus-safe, bus-unsafe, leguna-safe, and leguna-unsafe samples occur in both validation and test sets. Augmentation was applied only after splitting and only to training originals. Thresholds were tuned on validation predictions, while the 79-image test set remained untouched until final evaluation. The ensemble members (ResNet50 and ConvNeXt-Tiny) and all decision thresholds were fixed on the validation set before any test-set evaluation, so the test set was used exactly once and never for model selection, ensemble construction, or threshold tuning. The full nine-model test table is reported for transparency and did not drive the choice of the deployed model.

B. Model Configuration

The experiments were implemented in Python using scikit-learn for the classical pipelines [29], PyTorch for the neural model [30], and scikit-image for image processing [20]. A fixed random seed of 42 was used for reproducibility. Model and preprocessing settings are consolidated in Table II.

For HOG-based models, preprocessing and dimensionality-reduction steps were fitted on the training data and then applied to validation and test data. Deep models used normalized RGB image tensors rather than HOG descriptors.

TABLE II
MODEL CONFIGURATION

Model / Stage	**Input Representation**	**Key Configuration**
Image standardization	Source images	Orientation correction, RGB conversion, shorter side 512 px, center crop to 512×512, PNG output.
HOG feature extraction	Grayscale 128×128	9 orientation bins, 16×16 pixel cells, 2×2 cell blocks, L2-Hys normalization, 1764-d feature vector.
Logistic Regression	HOG features	StandardScaler and PCA (95% variance retained) on training data, then Logistic Regression (`lbfgs`, balanced class weights).
Gaussian Naive Bayes	HOG features	StandardScaler and PCA (95% variance retained) on training data, then Gaussian Naive Bayes.
SVM with RBF kernel	HOG features	StandardScaler and PCA (95% variance retained) on training data; RBF-kernel SVM with penalty C, kernel width γ, and kernel selected by 5-fold `GridSearchCV` (balanced class weights, probability estimates enabled).
CNN (from scratch)	RGB 128×128	Four convolutional blocks (32/64/128/256 channels) with batch normalization and ReLU, global average pooling, dropout, and a two-layer linear head; about 1.2M parameters.
ResNet18	RGB 224×224	ImageNet-pretrained residual network, two-phase fine-tuning.
EfficientNet-B0	RGB 224×224	ImageNet-pretrained compound-scaled network, two-phase fine-tuning.
ResNet50	RGB 224×224	Deeper ImageNet-pretrained residual network, two-phase fine-tuning, horizontal-flip TTA.
ConvNeXt-Tiny	RGB 224×224	ImageNet-pretrained modern convolutional network, two-phase fine-tuning, horizontal-flip TTA.
Ensemble	RGB 224×224	Probability-level average of ResNet50 and ConvNeXt-Tiny, with balanced, high-recall, and max-recall thresholds tuned on validation.
Deep-model training	RGB tensors	AdamW optimizer, cosine-annealing learning rate, two-phase schedule (head-only warm-up for about 5 epochs, then full-network fine-tuning), batch size 24–32, 45–70 epochs, label smoothing 0.05, dropout 0.4, weighted sampling, and class-weighted or focal loss for imbalance.
Evaluation protocol	Train/validation/test	Four-way stratified 70/15/15 split; 365 training originals expanded to 2,730 train images, with 79 unaugmented validation and 79 untouched test images.

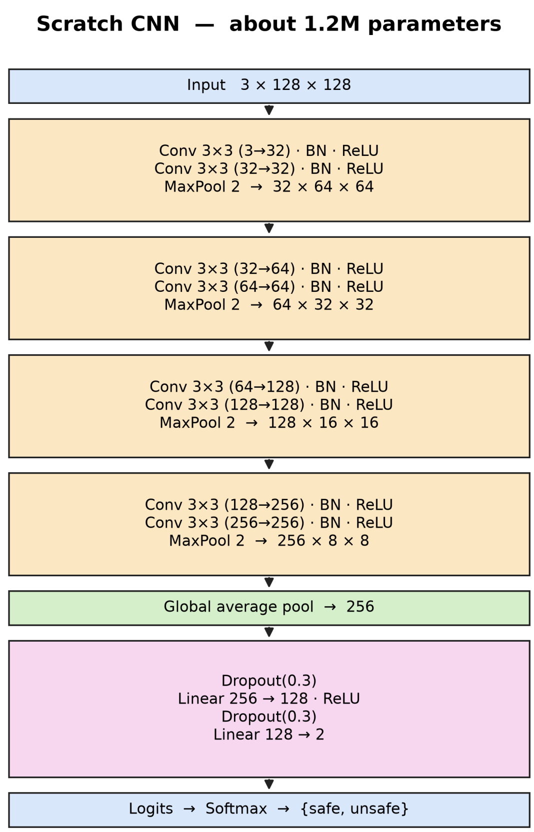


Fig. 6. Scratch CNN architecture used for image-level binary classification.

For neural-model inference, the output logits $\mathbf{z} = (z_0, z_1) \in \mathbb{R}^2$ are mapped to class probabilities by the softmax function,

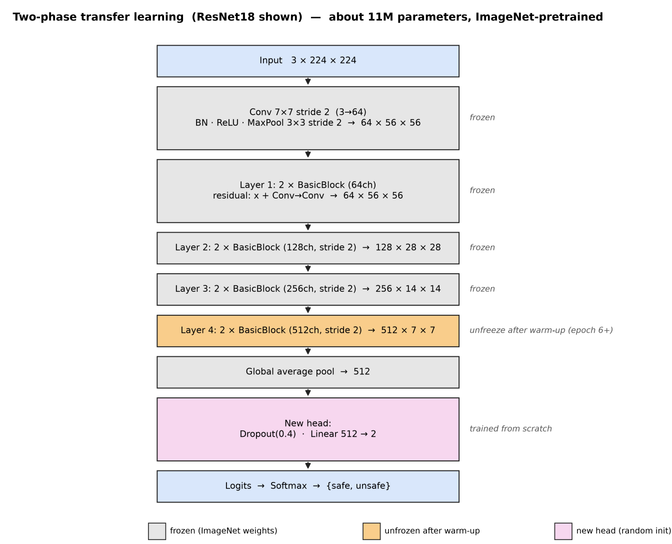


Fig. 7. Two-phase transfer-learning scheme, illustrated with ResNet18. The ImageNet-pretrained backbone is frozen during a short head-only warm-up, its deepest stage is then unfrozen for fine-tuning, and a new classifier head is trained from scratch. ResNet50, EfficientNet-B0, and ConvNeXt-Tiny follow the same procedure at a 224×224 input resolution.

$$p_k = \frac{\exp(z_k)}{\sum_{j=0}^{1} \exp(z_j)}, \qquad k \in \{0, 1\}, \tag{6}$$

so that p_1 is the estimated probability of the unsafe class. Neural models were trained with the AdamW optimizer, a decoupled weight-decay variant of Adam [31], under a cosine-annealing learning-rate schedule; the complete training configuration is listed in Table II.

C. Evaluation Metrics

Model performance was assessed using accuracy, balanced accuracy, precision, recall, specificity, the F_1-score, the Matthews correlation coefficient (MCC), ROC-AUC, and PR-AUC. Since the two classes are approximately balanced, both ROC-AUC and PR-AUC are reported to provide a more complete view of classification performance. Particular attention is given to recall for the unsafe/risky class, as failing to identify an unsafe case carries greater practical concern. Let TP, TN, FP, and FN denote the numbers of true positives, true negatives, false positives, and false negatives, respectively, where unsafe/risky hanging travel is treated as the positive class. The threshold-dependent evaluation metrics are defined as follows:

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN}, \tag{7}$$

$$\text{Precision} = \frac{TP}{TP+FP}, \qquad \text{Recall} = \frac{TP}{TP+FN}, \tag{8}$$

$$\text{Specificity} = \frac{TN}{TN+FP}, \tag{9}$$

$$F_1 = \frac{2\,\text{Precision}\cdot\text{Recall}}{\text{Precision}+\text{Recall}} = \frac{2\,TP}{2\,TP+FP+FN}. \tag{10}$$

Recall (equivalently, sensitivity or the true positive rate) quantifies unsafe-class coverage, whereas specificity (the true negative rate) quantifies safe-class coverage. Because a single decision threshold can favor one class, balanced accuracy reports the unweighted mean of the two per-class recalls and is therefore insensitive to class prevalence,

$$\begin{aligned}\text{Balanced Accuracy} &= \frac{\text{Recall}+\text{Specificity}}{2} \\ &= \frac{1}{2}\left(\frac{TP}{TP+FN}+\frac{TN}{TN+FP}\right).\end{aligned} \tag{11}$$

The Matthews correlation coefficient (MCC) measures the correlation between the predicted and true binary labels. By incorporating all four entries of the confusion matrix, it produces a high score only when the classifier performs consistently well across both classes. This makes MCC a particularly informative single-value measure, especially when class distributions are imbalanced [28]:

$$\text{MCC} = \frac{TP\cdot TN - FP\cdot FN}{\sqrt{(TP{+}FP)(TP{+}FN)(TN{+}FP)(TN{+}FN)}}. \tag{12}$$

The coefficient ranges from -1 to $+1$. A value of $+1$ represents perfect agreement between predictions and true labels, 0 indicates no correlation, and -1 reflects complete disagreement. When the denominator is undefined because one of its factors is zero, the coefficient is conventionally assigned a value of 0. In addition to these threshold-dependent measures, two threshold-independent metrics are used to assess ranking performance across all possible decision thresholds. ROC-AUC measures the area under the receiver operating characteristic curve, which plots the true positive rate against the false positive rate. It can also be interpreted as the probability that the model assigns a higher score to a randomly selected unsafe image than to a randomly selected safe image. PR-AUC, reported as average precision, summarizes the relationship between precision and recall across different classification thresholds.

$$\text{AP} = \sum_n (R_n - R_{n-1})\,P_n, \tag{13}$$

where P_n and R_n are the precision and recall at the n-th threshold; its no-skill baseline equals the unsafe-class prevalence, which makes it informative for the safety-critical positive class. Because missed unsafe cases have direct safety implications, unsafe-class recall is emphasized alongside overall accuracy, MCC, and AUC. Given the small test set, uncertainty is reported as 95% Wilson score intervals, and pairwise differences between the strongest models are assessed with McNemar's exact test.

VI. Results

A. Quantitative Comparison

Table III reports balanced operating-point performance on the untouched 79-image test set. The ResNet50–ConvNeXt ensemble reaches 0.861 accuracy, 0.947 unsafe recall, and 0.970 ROC-AUC. ResNet50 attains the highest single-model accuracy (0.924) and MCC (0.848), while EfficientNet-B0 attains the highest unsafe recall (0.974) and ROC-AUC (0.979). All nine models are compared in Fig. 8.

B. ROC and AUC Analysis

AUC values are reported in Table III. Figure 9a shows held-out ROC curves for all reported models, and Fig. 9b shows the corresponding precision–recall curves.

C. Error Analysis

False negatives are the most safety-critical error type because they correspond to unsafe/risky hanging-travel cases predicted as safe/normal. At the balanced operating point, the ensemble detects 94.7% of unsafe test images, missing only 2 of 38 hangers. Because the retrained dataset is close to balanced, the tuned balanced, high-recall, and max-recall thresholds fall within 0.006 of one another, so all three settle at the same operating point of 86.1% accuracy and 94.7% unsafe recall. The earlier regime, in which perfect recall could only be bought at low accuracy, no longer appears: the additional unsafe training data lets the ensemble reach high recall without sacrificing accuracy. Table IV lists the three modes, and Fig. 10 shows the threshold trade-off.

TABLE III
HELD-OUT TEST PERFORMANCE FOR ALL REPORTED MODELS

Model	Acc.	Bal. Acc.	Unsafe R	Unsafe P	Spec.	F_1	AUC	PR-AUC	MCC
Ensemble	0.861	0.864	0.947	0.800	0.780	0.867	0.970	0.978	0.734
ResNet50	**0.924**	**0.924**	0.921	**0.921**	**0.927**	**0.921**	0.967	0.975	**0.848**
ConvNeXt-Tiny	0.835	0.840	0.947	0.766	0.732	0.847	0.972	0.979	0.691
EfficientNet-B0	0.835	0.841	**0.974**	0.755	0.707	0.851	**0.979**	**0.980**	0.701
ResNet18	0.911	0.912	0.921	0.897	0.902	0.909	0.971	0.976	0.823
CNN	0.861	0.861	0.868	0.846	0.854	0.857	0.929	0.896	0.722
SVM (RBF)	0.873	0.874	0.895	0.850	0.854	0.872	0.952	0.958	0.748
Logistic Regression	0.810	0.810	0.816	0.795	0.805	0.805	0.887	0.900	0.620
Naive Bayes	0.734	0.734	0.737	0.718	0.732	0.727	0.866	0.884	0.468

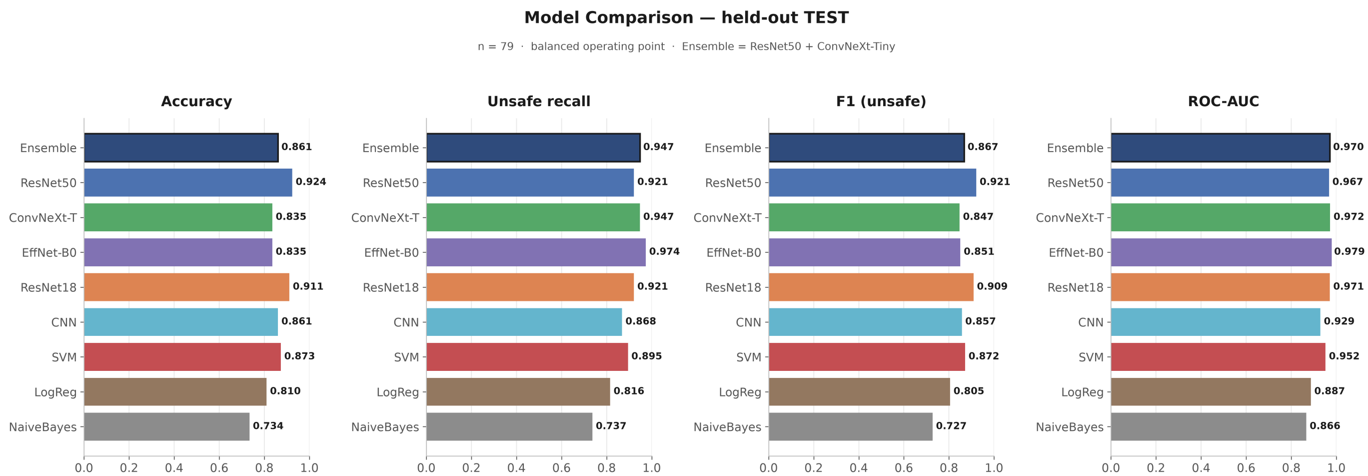


Fig. 8. Held-out test comparison of accuracy, unsafe recall, unsafe F_1, and ROC-AUC across all nine reported models.

TABLE IV
SELECTABLE OPERATING POINTS FOR THE DEPLOYED ENSEMBLE. ON THE NEAR-BALANCED RETRAINED DATA THE THREE TUNED THRESHOLDS NEARLY COINCIDE, SO THE MODES REPORT THE SAME TEST OPERATING POINT.

Mode	Threshold	Test Acc.	Unsafe R	Unsafe P	Spec.
Balanced	0.170	0.861	0.947	0.800	0.780
High recall	0.175	0.861	0.947	0.800	0.780
Max recall	0.175	0.861	0.947	0.800	0.780

VII. DISCUSSION

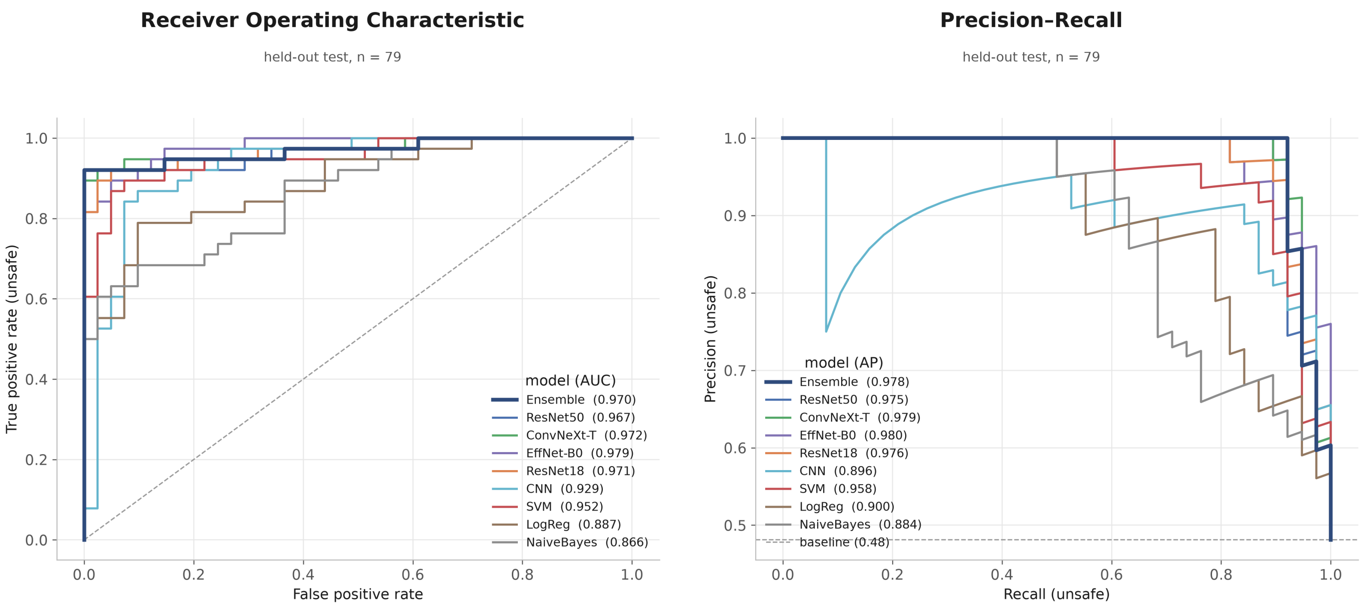
A. Interpretation of Results

The retrained ensemble provides a strong, recall-oriented balanced operating point, reaching 86.1% test accuracy, 94.7% unsafe recall, 0.867 unsafe F_1, 0.970 ROC-AUC, and 0.978 PR-AUC. The additional unsafe training data raises AUC beyond the previous 0.949 result and substantially improves balanced unsafe recall.

Among individual deep models, ResNet50 leads on accuracy (92.4%) with balanced 92.1% unsafe precision and recall, while EfficientNet-B0 reaches the highest unsafe recall (97.4%) and the top ROC-AUC (0.979). Among classical models, SVM remains the strongest HOG baseline, reaching 87.3% accuracy, 89.5% unsafe recall, and 0.952 AUC.

We deploy the ResNet50–ConvNeXt ensemble rather than the single most accurate model because the two error types carry asymmetric cost: a false negative, a missed hanging passenger, is more consequential than a false alarm. The ensemble attains 94.7% unsafe recall, above ResNet50's 92.1%, and averaging two backbones yields probabilities that are more stable across the operating thresholds than any single network. EfficientNet-B0 reaches an even higher unsafe recall (97.4%) and is a strong recall-oriented alternative, but its lower balanced accuracy (83.5%) produces more false alarms, so the ensemble was retained for its better accuracy–recall balance.

Gaussian Naive Bayes underperforms because its conditional-independence assumption is poorly matched to HOG descriptors. Neighboring gradient bins and block-normalized features are correlated by construction, so treating them as independent weakens the model's ability to distinguish visually similar safe/normal and unsafe/risky



(a) ROC curves.

(b) Precision–recall curves.

Fig. 9. Receiver operating characteristic and precision–recall curves comparing all nine models on the held-out test set.

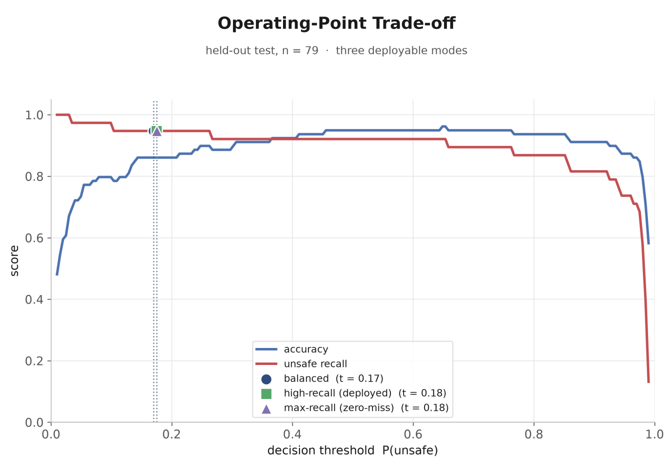


Fig. 10. Ensemble threshold selection on the held-out test set: accuracy and unsafe recall against the decision threshold, with the balanced, high-recall, and max-recall operating points marked. Because the retrained dataset is near-balanced, the three points nearly coincide.

scenes.

The train-to-test gaps in Table V range from 6.7 to 16.5 percentage points. Logistic Regression has the smallest gap, while ConvNeXt-Tiny has the largest. These gaps are markedly smaller than in the previous run, which indicates that the expanded set of unique originals reduces overfitting, even though it does not remove it entirely. Figure 11 visualizes the per-model train, validation, and test error and the train-to-test gap.

For this application, false negatives are more consequential than ordinary classification errors because they correspond to missed risky travel conditions. Precision and overall accuracy remain relevant for reducing false alarms, but a safety-oriented monitoring system should prioritize reducing missed unsafe cases while maintaining acceptable discrimination across normal and risky scenes.

B. Limitations

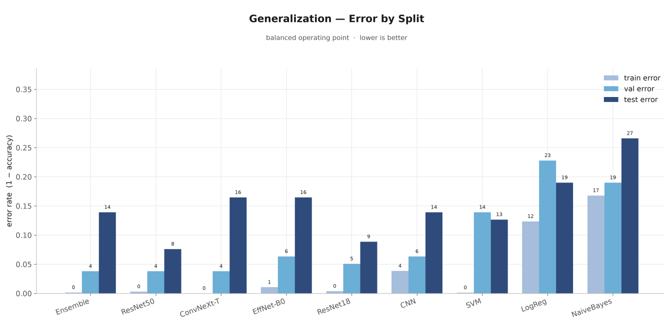
The main limitation of this study is the limited number of unique source images. The updated dataset contains 523 labeled originals with 251 unsafe cases, which is still not enough to fully cover the diversity of vehicle layouts, camera viewpoints, passenger postures, weather, illumination, and traffic environments encountered in deployment. With only 79 test images, individual metrics carry wide uncertainty: a 95% Wilson score interval on the ensemble's 86.1% accuracy spans $[76.8, 92.0]\%$. A McNemar exact test finds no significant difference among the strongest models (for example, ResNet50 versus the ensemble gives $p = 0.125$ and ResNet50 versus ResNet18 gives $p = 1.0$), so these models are statistically indistinguishable on this test set, and the per-metric maxima in Table III should be read as descriptive rather than as proof of a single definitively best model. Because nine models are compared on one split, the top scores are also subject to selection ("winner's-curse") optimism, and the reported interval captures only test-set sampling noise, not train/validation-split or random-seed variability. Cross-validated or repeated-seed evaluation would tighten these estimates and is left to future work.

The neural models also depend on training-time augmentation to increase effective sample diversity. Augmentation makes the models less sensitive to controlled transformations, but it does not replace additional real images containing natural occlusion, motion blur, dense crowding, and non-standard bus or leguna structures. These factors can make unsafe/risky positive cases difficult to distinguish from visually cluttered safe/normal scenes.

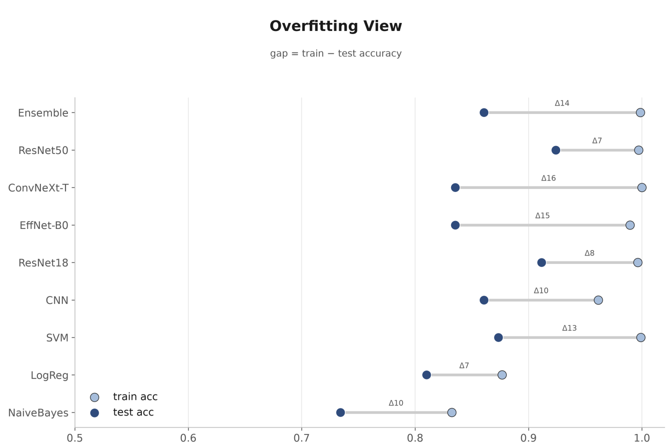
The current system is limited to image-level binary classification. It does not localize hanging passengers, estimate crowd density, track passengers across frames, or evaluate performance on large-scale real-time video streams. Roadside transport images may also include identifiable people, license plates, and location cues, so deployment would require anonymization and data-governance safeguards. The method should therefore be interpreted as a baseline for visual safety classification rather than an operational monitoring system.

TABLE V
TRAIN, VALIDATION, AND TEST GENERALIZATION

Model	Train Acc.	Train Err.	Val Acc.	Val Err.	Test Acc.	Test Err.	Test R
Ensemble	0.999	0.001	0.962	0.038	0.861	0.139	0.947
ResNet50	0.997	0.003	0.962	0.038	0.924	0.076	0.921
ConvNeXt-Tiny	1.000	0.000	0.962	0.038	0.835	0.165	0.947
EfficientNet-B0	0.989	0.011	0.937	0.063	0.835	0.165	0.974
ResNet18	0.996	0.004	0.949	0.051	0.911	0.089	0.921
CNN	0.962	0.038	0.937	0.063	0.861	0.139	0.868
SVM (RBF)	0.999	0.001	0.861	0.139	0.873	0.127	0.895
Logistic Regression	0.877	0.123	0.772	0.228	0.810	0.190	0.816
Naive Bayes	0.832	0.168	0.810	0.190	0.734	0.266	0.737



(a) Train, validation, and test error.



(b) Train-to-test generalization gap.

Fig. 11. Generalization behavior of all nine models after retraining.

VIII. CONCLUSION AND FUTURE WORK

This paper developed a computer-vision and machine-learning pipeline for detecting unsafe hanging travel in Dhaka public transport. The task was formulated as binary image classification over safe/normal and unsafe/risky passenger conditions. A localized bus and leguna dataset was prepared through cleaning, annotation, preprocessing, and augmentation, with HOG features used for classical models and RGB inputs used for CNN-based classification.

The updated result set compares traditional machine learning algorithms like Logistic Regression, Gaussian Naive Bayes,with deep networks like SVM with an RBF kernel, a CNN trained from scratch, ResNet18, EfficientNet-B0, ResNet50, ConvNeXt-Tiny, and a ResNet50–ConvNeXt ensemble. On the untouched test set, the deployed ensemble reaches a recall-oriented balanced result with 86.1% accuracy, 94.7% unsafe recall, and 0.970 ROC-AUC, while ResNet50 gives the highest single-model accuracy at 92.4%. Because the retrained data is near-balanced, the balanced, high-recall, and max-recall thresholds coincide at the same 86.1%-accuracy and 94.7%-recall operating point rather than trading accuracy for a separate perfect-recall mode.

As a practical outcome of this work, we developed an online platform that classifies public-transport images as either safe/normal or unsafe/risky. The project resources are publicly accessible through the following links: the source-code repository, the project dataset, and the deployed web platform.

Future work should focus on collecting a larger localized dataset, improving recall for unsafe hanging travel, evaluating the system on real-time video streams, and designing privacy-aware deployment procedures. Detection-based models such as YOLO [6] should also be tested to localize hanging passengers rather than only classify whole images. License-plate recognition could be incorporated in future work to identify vehicles after an unsafe condition has been detected.

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